



Developing an Explainable Artificial Intelligence System for the Mobile-Based Diagnosis of Febrile Diseases Using Random Forest, LIME, and GPT

Kingsley F. Attai¹, Constance Amannah², Moses Ekpenyong^{3,4}, Daniel E. Asuquo⁵, Oryina K. Akputu⁶, Okure U. Obot³, Peterben C. Ajuga⁷, Jeremiah C. Obi⁷, Omosivie Maduka⁸, Christie Akwaowo⁹, Faith-Michael Uzoka¹⁰

¹Department of Mathematics and Computer Science, Ritman University, Ikot Ekpene, Nigeria

²Department of Computer Science, Ignatius Ajuru University of Education, Port Harcourt, Nigeria

³Department of Computer Science, Faculty of Computing, University of Uyo, Uyo, Nigeria

⁴STEM Centre, University of Uyo and Centre for Research, University of Uyo, Uyo, Nigeria

⁵Department of Information Systems, Faculty of Computing, University of Uyo, Uyo, Nigeria

⁶Department of Computing Sciences, Admiralty University of Nigeria, Ibusa, Nigeria

⁷Novena Computers and Technologies Limited, Uyo, Nigeria

⁸University of Port Harcourt Teaching Hospital, Port Harcourt, Nigeria

⁹University of Uyo Teaching Hospital, Uyo, Nigeria

¹⁰Department of Mathematics and Computing, Mount Royal University, Calgary, Canada

Objectives: This study proposes a mobile-based explainable artificial intelligence (XAI) platform designed for diagnosing febrile illnesses. **Methods:** We integrated the interpretability offered by local interpretable model-agnostic explanations (LIME) and the explainability provided by generative pre-trained transformers (GPT) to bridge the gap in understanding and trust often created by machine learning models in critical healthcare decision-making. The developed system employed random forest for disease diagnosis, LIME for interpretation of the results, and GPT-3.5 for generating explanations in easy-to-understand language. **Results:** Our model demonstrated robust performance in detecting malaria, achieving precision, recall, and F1-scores of 85%, 91%, and 88%, respectively. It performed moderately well in detecting urinary tract and respiratory tract infections, with precision, recall, and F1-scores of 80%, 65%, and 72%, and 77%, 68%, and 72%, respectively, maintaining an effective balance between sensitivity and specificity. However, the model exhibited limitations in detecting typhoid fever and human immunodeficiency virus/acquired immune deficiency syndrome, achieving lower precision, recall, and F1-scores of 69%, 53%, and 60%, and 75%, 39%, and 51%, respectively. These results indicate missed true-positive cases, necessitating further model fine-tuning. LIME and GPT-3.5 were integrated to enhance transparency and provide natural language explanations, thereby aiding decision-making and improving user comprehension of the diagnoses. **Conclusions:** The LIME plots revealed key symptoms influencing the diagnoses, with bitter taste in the mouth and fever showing the highest negative influence on predictions, and GPT-3.5 provided natural language explanations that increased the reliability and trustworthiness of the system, promoting improved patient outcomes and reducing the healthcare burden.

Keywords: Natural Language Processing, Artificial Intelligence, Machine Learning, Telemedicine, Mobile Applications

Submitted: January 10, 2025, **Revised:** April 18, 2025, **Accepted:** April 23, 2025

Corresponding Author

Moses Ekpenyong

Department of Computer Science, University of Uyo, Ikpa/Nwaniba Road, Uyo, Akwa Ibom State 520003, Nigeria. Tel: +234-8037933961, E-mail: mosesekpenyong@uniuyo.edu.ng (<https://orcid.org/0000-0001-6774-5259>)

This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<http://creativecommons.org/licenses/by-nc/4.0/>) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

I. Introduction

Febrile diseases such as malaria, typhoid fever, human immunodeficiency virus (HIV) and acquired immune deficiency syndrome (AIDS), tuberculosis, respiratory tract infection, and urinary tract infection remain significant public health challenges, particularly in resource-scarce settings. These resource-constrained areas frequently lack adequate access to medical facilities and qualified healthcare professionals, resulting in delays in diagnosis and treatment. The diagnostic process is further complicated by overlapping symptoms shared among these diseases [1], underscoring the need for timely and accurate diagnoses to ensure appropriate treatments, avoid complications, and minimize disease spread or transmission risks.

Mobile health (mHealth) applications have emerged as a promising avenue for enhancing healthcare delivery and facilitating timely diagnostic services [2,3]. Artificial intelligence (AI) and machine learning (ML) have significantly improved diagnostic efficiency and accuracy in medical systems [4], and these techniques have been specifically applied in diagnosing febrile diseases [5-10].

However, the “black box” nature of ML models introduces challenges related to clarity and transparency, reducing the interpretability and comprehensibility of the underlying decision logic. In healthcare contexts, this opacity raises concerns about the reliability and transparency of diagnostic systems [11]. To address issues related to transparency and trust, explainable AI (XAI) approaches have emerged to improve interpretability of ML models [12,13]. The local interpretable model-agnostic explanations (LIME) [14] method is an XAI approach that locally approximates complex models with simpler, interpretable ones. Large language models (LLMs), such as generative pre-trained transformers (GPT), have demonstrated an enhanced ability to understand and produce human-like text [15]. Such models hold considerable potential for improving comprehension among lay healthcare users by translating complex diagnostic results into easily understandable language. This capability is particularly important in mHealth applications, where clear comprehension of diagnostic data is crucial for recommending appropriate treatment plans.

Although ML models are widely used for illness diagnosis, medical professionals often find it difficult to trust their predictions due to the limited explainability associated with generated results. Moreover, despite the successful human-like text generation by LLMs, medical diagnostics have not fully embraced their potential. Current diagnostic tools typi-

cally employ a single methodological approach rather than combining multiple methods.

This paper integrates the interpretability of LIME, the explainability of GPT-3.5, and the predictive capabilities of random forest (RF) into a mobile-based diagnostic application aimed at improving febrile disease diagnosis. By incorporating LIME, this study clarifies diagnostic predictions by visualizing the relative importance of each symptom, thereby enabling healthcare professionals to identify symptoms that most significantly influence predicted health conditions. The results are further enhanced by GPT-3.5, which provides a user-friendly interface by translating technical explanations into easily comprehensible language for non-specialists. In resource-scarce regions, the improved application will aid healthcare workers in diagnosing patient conditions rapidly and accurately, potentially leading to better health outcomes.

The remainder of this paper is organized as follows: Section II describes the methodology, presenting the proposed system framework, dataset description, preprocessing steps, diagnostic methods, and interpretability models designed to improve diagnostic explainability. Section III presents the study's findings, evaluating the effectiveness of ML models, demonstrating how XAI provides insights into model decisions, and detailing the development of the mobile-based XAI system. Section IV concludes the study, identifies its limitations, and provides recommendations for future research directions and enhancements in diagnostic methodologies.

II. Methods

1. Proposed System Framework

A mobile-based diagnostic system for identifying febrile diseases was developed using agile software development methods, emphasizing iterative user interactions. The system is designed to be user-centered and flexible, facilitating ongoing enhancements based on user feedback and real-time data generation and transmission. The architecture of the proposed system is presented in Figure 1. It comprises key components including medical experts, healthcare workers, a diagnostic system, and a mobile device used for collecting patient information, vital signs, and symptoms. These data are analyzed and stored both locally and on the cloud to aid decision-making. Medical professionals provided patient data, which were subsequently preprocessed to ensure suitability for ML and LLM modeling.

The RF algorithm, which forms the basis of this tool, was selected due to its robustness in medical diagnosis and its

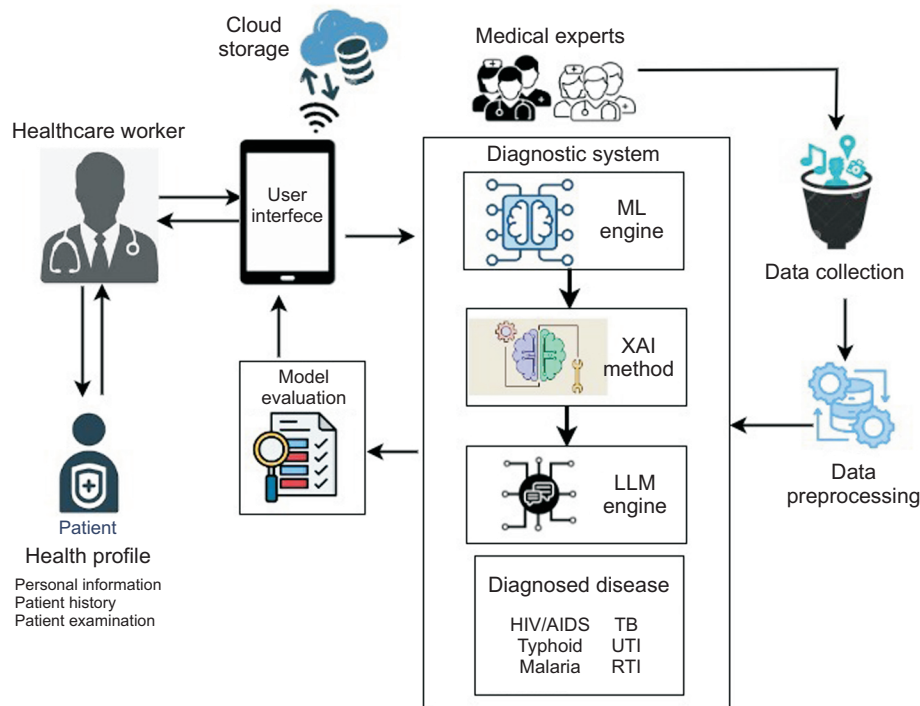


Figure 1. Architecture of the explainable artificial intelligence (XAI) febrile disease diagnosis system. ML: machine learning, LLM: large language model, HIV: human immunodeficiency virus, AIDS: acquired immune deficiency syndrome, TB: tuberculosis, UTI: urinary tract infection, RTI: respiratory tract infection.

resilience when working with complex datasets. Trained on an extensive dataset of clinical symptoms associated with febrile illnesses, the algorithm predicts the likelihood of febrile diseases based on patient inputs. To address interpretability concerns inherent in ML models, the system integrates LIME, which generates understandable explanations for the predictions made by the RF model, highlighting the symptoms most influential in each disease diagnosis.

Furthermore, GPT-3.5 was employed by the system to provide natural-language explanations of diagnostic results obtained from LIME. This feature ensures that users, regardless of their medical expertise, clearly understand the diagnostic outcomes. The iterative integration of GPT-3.5 and other system features was facilitated by agile development practices, allowing rapid adaptations to evolving user needs and requirements. Because the diagnostic tool is implemented as a mobile application, users can access it conveniently in diverse settings, especially in resource-scarce areas.

2. Dataset Description and Data Preprocessing

This study utilized 4,870 patient records from the New Frontiers in Research Fund project [16]. The dataset contained demographic information, patient symptoms, risk factors, suspected diagnoses, further investigations, and confirmed diagnoses. Symptom severity was rated using a 5-point scale (5 = very severe, 4 = severe, 3 = moderate, 2 = mild, and 1 = absent). Additionally, the dataset included susceptibility

to non-clinical risk factors and medical experts' suspected diagnoses. Further investigations, such as blood film examination, serology, full blood counts, etc., were conducted before confirmed diagnoses were established. The severity of both suspected and confirmed diagnoses was rated on a linguistic scale (6 = very high; 5 = high; 4 = moderate; 3 = low; 2 = very low; and 1 = absent). The dataset included the following diseases: malaria, HIV/AIDS, typhoid fever, tuberculosis, dengue fever, urinary tract infection, yellow fever, respiratory tract infection, and Lassa fever. During data preprocessing, records containing missing symptoms, as well as diseases outside the scope of this study (yellow fever, dengue fever, and Lassa fever), were removed to maintain dataset integrity. Patients under 5 years old were also excluded from the study because the data collection instrument did not account for symptoms in children younger than five, who are unable to clearly express certain symptoms. The dataset was then narrowed to include only symptoms and confirmed diagnoses relevant to the study, removing columns related to physicians' suspected diagnoses. Disease labels (output labels) were converted into binary encoding—absent “0” (originally 1 = absent) and present “1” (originally 6 = very high; 5 = high; 4 = moderate; 3 = low; and 2 = very low). This simplification facilitated classification tasks, reduced model complexity, and improved performance, as the model no longer needed to distinguish among multiple overlapping classes. After preprocessing, the dataset consisted of 3,914

records. Table 1 lists the 32 symptoms and 6 confirmed diagnoses used in the study.

Relevant medical experts and healthcare workers were involved throughout the iterative development process to ensure comprehensive understanding of the proposed system's features. Several meetings with stakeholders were held to solicit feedback on system functionality. The elicited requirements were then organized and documented in standard software engineering formats. Figure 2 illustrates the use case diagram, providing a graphical representation of interactions among users, healthcare workers, patients, and administrators.

3. Diagnostic, Interpretability, and Explainability Models

The study utilized Visual Studio Code and Google Colab, along with core Python libraries and packages including matplotlib, sklearn, numpy, pandas, seaborn, flask, flask-sqlalchemy, flet, joblib for model loading, an interpreter, automator, and prompt generator. The prompt generator organizes the list of diagnosed diseases and symptoms generated by the LIME model into structured prompts and saves these prompts in JSON format, which are then sent as requests to the ChatGPT API, with responses subsequently saved. The system consists of two main components: a frontend created with Flet that visualizes data using Matplotlib and Seaborn; and a backend that utilizes Flask for MySQL database management and API integration via Flask-SQLAlchemy. The diagnostic model, including performance metrics, was developed using the RF algorithm with hyperparameter tuning through GridSearchCV to enhance diagnostic precision. RF is an ensemble ML technique that improves predictive accuracy by combining multiple decision trees, effectively recognizing patterns in high-dimensional and complex problems [17]. The RF algorithm was selected specifically for its robustness in medical diagnosis contexts and resilience when working with complicated datasets.

LIME generated visual explanations showing how symptom features influenced diagnoses. It provided local interpretability by approximating the complex model with a simpler, interpretable model around a specific prediction, thereby making the model's reasoning transparent to healthcare workers. LIME identified the symptoms with the most significant influence on model decision-making, offering concise, locally interpretable explanations for individual diagnoses. GPT converted these explanations into plain, natural language, facilitating healthcare workers' understanding and trust in the model's predictions. The combination of these methods not only enhanced diagnostic performance but also improved

Table 1. Symptoms and diseases used in the study

Symptom/disease	Abbreviation
Abdominal pains	ABDPN
Bitter taste in mouth	BITAIM
Bloody urine	BLDYURN
Catarrh	CTRH
Chest indraw	CHSIND
Chest pain	CHSPN
Chills and rigors	CHLNRI
Constipation	CNST
Cough (initial dry)	CGHTRY
Difficulty breathing	DIFBRT
Dry cough	DRYCGH
Fatigue	FTG
Fever	FVR
High-grade fever	HGGDFVR
Stepwise rise fever	SWRFVR
Low-grade fever	LWGDFVR
Foul breath	FOLBRT
Generalized body pain	GENBDYPN
Generalized rashes	GENRSH
Headaches	HDACH
Lethargy	LTG
Lymph node swelling	LMPNDSWL
Muscle and body pain	MSCBDYPN
Mouth ulcer	MUTUCR
Nausea	NUS
Night sweats	NGTSWT
Painful urination	PNFLURNTN
Sore throat	SRTRT
Suprapubic pains	SPPBPN
Urinary frequency	URNFQC
Vomiting	VMT
Wheezing	WHZ
Malaria	MAL
Typhoid/Enteric fever	ENFVR
HIV and AIDS	HVAD
Urinary tract infection	UTI
Respiratory tract infection	RTI
Tuberculosis	TB

HIV: human immunodeficiency virus, AIDS: acquired immune deficiency syndrome.

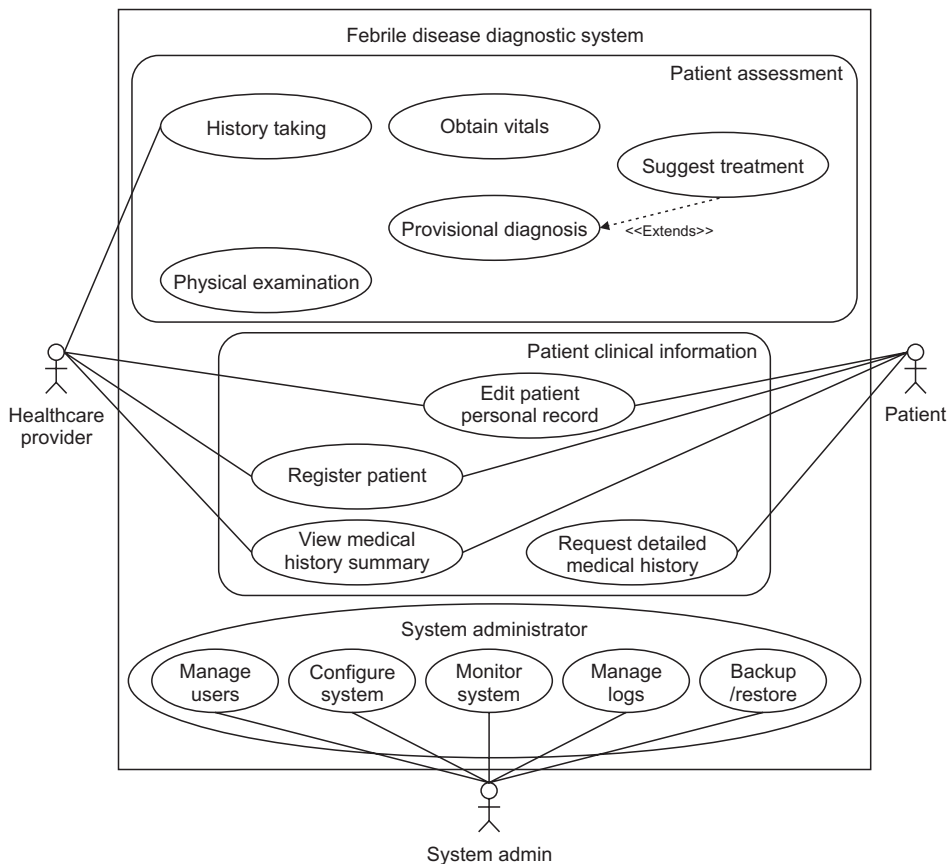


Figure 2. Use case diagram of the diagnostic system.

user confidence by clearly and understandably communicating the rationale behind each diagnosis. The sample prompt, illustrated in Table 2, leveraged the list of diagnosed diseases and patient symptoms generated by the LIME model, alongside explanations regarding the influence of each symptom on the diagnosis. Before submission to the ChatGPT API, the generated prompt was stored in a variable and formatted into JSON.

III. Results

This study evaluated the performance of three diagnostic models—multilayer perceptron (MLP), extreme gradient boosting (XGBoost), and RF—across six disease categories: MAL (Malaria), ENFVR (typhoid/enteric fever), HVAD (HIV and AIDS), UTI (urinary tract infection), RTI (respiratory tract infection), and TB (tuberculosis). Unlike deep learning models, which typically require large training datasets to achieve strong performance and avoid overfitting, these traditional machine learning models generally demonstrate effective generalization on smaller datasets and are less computationally demanding. The selected models were evaluated using standard classification metrics, specifically precision, recall, and F1-score. These metrics were chosen

because they effectively assess diagnostic models, especially in clinical contexts where false negatives and false positives have significant implications. Table 3 summarizes the comparative results, highlighting the strengths and limitations of each model in accurately diagnosing the respective conditions. Among the evaluated models, RF outperformed MLP and XGBoost across most disease categories. In healthcare settings, RF is particularly effective with multi-label and imbalanced data [18], making it especially suitable for diagnostic tasks where certain conditions, such as HVAD and TB, may be underrepresented. Its ensemble learning strategy, which aggregates predictions from multiple decision trees, enhances generalization and mitigates overfitting, contributing to its robust and reliable performance.

The performance of the RF model in diagnosing each disease is further visualized in Figure 3. The model demonstrated strong performance in detecting malaria, achieving precision, recall, and F1-scores of 85%, 91%, and 88%, respectively. However, its performance was notably weaker for typhoid fever and HIV/AIDS, with F1-scores of 39% and 51%, respectively, indicating a tendency to miss true-positive cases of these diseases. For urinary tract infections and respiratory tract infections, the model showed balanced performance, achieving F1-scores of 72% for both conditions. The

Table 2. Sample prompt of diagnosed diseases and patient symptoms

Explain the LIME results below (disease or diseases and significant symptoms) to a physician with not more than 300 words: Predictions ['Typhoid Fever Likely', 'HIV/AIDS Likely', 'Urinary Tract Infection Likely', 'Respiratory Tract Infection Likely', 'Tuberculosis Likely'] LIME Explanation; BITTER TASTE IN MOUTH <= 1.00; -0.16758919765052108 Painful Urination > 1.00; -0.06605868111597245 Suprapubic_Pain > 1.00; -0.06550467141471979 Difficulty Breathing <= 1.00; 0.06071678705925383 Wheezing <= 1.00; 0.0590462921972035 Headache <= 1.00; -0.05826547312462451 CHILLS AND RIGORS <= 1.00; -0.05630271637802242 CHEST INDRAW <= 1.00; 0.04070095788472542 Generalized Body Pain > 3.00; 0.03974979336103006 1.00 < CATARRH <= 2.00; 0.03816070620416529 Urinary_Frequency > 1.00; -0.03770207672685268 ABDOMINAL PAIN > 3.00; -0.033902377938511155 2.00 < Fever <= 3.00; 0.032999053123995245 1.00 < Muscle and Body Pain <= 3.00; 0.03295097711760973 HGGDFever > 3.00; 0.032768000129691194 BLOODY URINE > 1.00; -0.029955841393233477 Cough (Initial Dry) <= 1.00; 0.026364166193042264 CHEST PAIN <= 1.00; 0.02354889167461192 Sore_Throat > 1.00; -0.021402483028506374 Lymph Node Swelling > 1.00; -0.0202552321969518 Vomiting > 2.00; 0.019057721531348118 Lethargy > 2.00; -0.017371438160508137 Mouth Ulcer > 1.00; -0.015856144250446513 Fatigue > 3.00; -0.015122895567807396 Generalized Rash <= 1.00; 0.014081682731125898 Foul Breath <= 1.00; 0.01260963643914623 1.00 < LWGDFever <= 3.00; 0.012187835432654306 CONSTIPATION <= 1.00; -0.012145436356090885 Night Sweat > 1.00; -0.010556292961671602 SWRFever > 2.00; -0.007451681601225083 Nausea > 2.00; -0.00406587833157149 Dry Cough > 1.00; -0.003064108573649665

detection of tuberculosis revealed moderate precision (77%) but lower recall (49%), resulting in an F1-score of 60%. These results indicate that although the RF model effectively diagnoses certain diseases, such as malaria, further data enrichment and model optimization are necessary to improve diagnostic performance across all disease categories.

Figure 4 presents a feature importance plot generated using the LIME framework applied to a random forest model. It highlights the most influential symptoms contribut-

ing to each disease diagnosis, such as difficulty breathing (DIFBRT), painful urination (PNFLURTN), and wheezing (WHZ). The least influential symptoms, including bitter taste in the mouth (BITAIM), fever (FVR), and generalized body pain (GENBDYPN), are also indicated. This visualization demonstrates that different diseases have distinct feature-importance profiles, underscoring LIME’s effectiveness in identifying symptom patterns unique to specific conditions. Such insights enable medical professionals to make

Table 3. Performance of the diagnostic models

		MAL	ENFVR	HVAD	UTI	RTI	TB
MLP	Precision	0.84	0.65	0.75	0.77	0.75	0.59
	Recall	0.87	0.42	0.34	0.65	0.63	0.59
	F1-score	0.85	0.51	0.46	0.70	0.69	0.64
XGBoost	Precision	0.84	0.64	0.62	0.77	0.77	0.72
	Recall	0.90	0.56	0.40	0.63	0.68	0.60
	F1-score	0.87	0.60	0.48	0.70	0.72	0.65
RF	Precision	0.85	0.69	0.75	0.80	0.77	0.77
	Recall	0.91	0.53	0.39	0.65	0.68	0.49
	F1-score	0.88	0.60	0.51	0.72	0.72	0.60

MLP: multilayer perceptron, XGBoost: extreme gradient boosting, RF: random forest, MAL: Malaria, ENFVR: Typhoid/Enteric fever, HVAD: human immunodeficiency virus (HIV) and acquired immune deficiency syndrome (AIDS), UTI: urinary tract infection, RTI: respiratory tract infection, TB: tuberculosis.

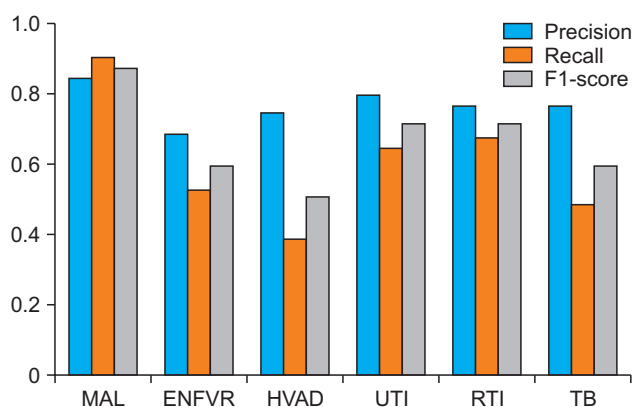


Figure 3. Performance evaluation of the random forest model. MAL: Malaria, ENFVR: Typhoid/Enteric fever, HVAD: human immunodeficiency virus (HIV) and acquired immune deficiency syndrome (AIDS), UTI: urinary tract infection, RTI: respiratory tract infection, TB: tuberculosis.

informed clinical decisions by understanding not only the model's predictions but also the rationale behind these predictions.

The febrile disease diagnostic system is compatible with Android OS version 4.0 and above and requires at least 2 GB RAM, 8 GB of minimum ROM storage, a portrait-oriented display layout, and an internet connection. The mobile application includes a user authentication screen shown in Figure 5A, and the primary dashboard for healthcare workers shown in Figure 5B. This dashboard enables healthcare professionals to view patient lists, register new patients, schedule appointments, and respond to patient requests. After registering a patient, healthcare workers access the patient dashboard displayed in Figure 5C. Through

the patient dashboard, healthcare workers can record vital signs, perform clinical examinations, take medical history (Figure 5D), and carry out provisional diagnoses (Figure 5E). The provisional diagnosis screen presents the patient's likely diseases, visualizes how symptoms contribute to each diagnosis, and includes GPT-3.5-generated explanations of diagnostic outcomes. Figure 5E also displays the output from the ChatGPT platform, highlighting both significant and less significant symptom contributions with clear, interpretable explanations.

IV. Discussion

The findings of this study illustrate the advantages and limitations of employing a mobile XAI system combining RF, LIME, and GPT-3.5 to diagnose febrile diseases. While the system exhibits impressive diagnostic accuracy for certain conditions, it struggles with others, highlighting specific areas for improvement. Firstly, the model's robustness in identifying common febrile illnesses was demonstrated through its high performance in detecting malaria, achieving precision, recall, and F1-scores of 85%, 91%, and 88%, respectively. This performance indicates that the system can effectively detect malaria cases, thus assisting healthcare workers in managing a disease that continues to significantly contribute to morbidity, particularly in resource-limited regions. The model notably minimizes false negatives, ensuring accurate identification of most true cases, which is crucial for timely medical intervention, as indicated by the high sensitivity in malaria detection.

The system demonstrated balanced performance, achiev-

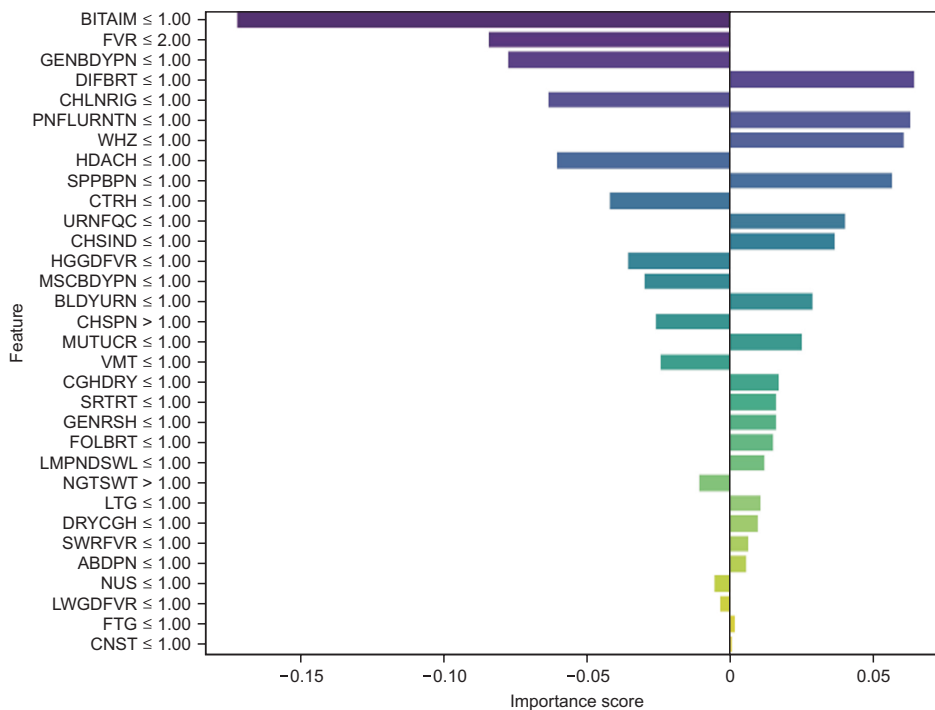


Figure 4. Feature importance plot showing locally interpretable, model-agnostic explanations (LIME) of the top predictive features.

ing F1-scores of 72% for both RTI and UTI, thus showing moderate effectiveness in these diagnostic categories. These results suggest that while the model functions adequately, further enhancements could be made, particularly in recall rates (68% for RTI and 65% for UTI). The findings indicate that some true-positive cases may be missed, potentially due to symptom overlaps between these diseases and others within the dataset. The utility of the model in identifying common diseases in resource-constrained settings could be further improved by fine-tuning feature importance and symptom weighting, thus increasing sensitivity for these conditions. Conversely, the model exhibited limited performance in detecting typhoid fever and HIV/AIDS, with precision, recall, and F1-scores of 69%, 53%, and 60% for typhoid fever, and 75%, 39%, and 51% for HIV/AIDS, respectively. The low recall rates suggest a significant number of actual cases, particularly for HIV/AIDS, were not detected. Misclassification may arise due to overlapping symptoms shared with other febrile diseases, potentially explaining the model's weaker performance. Additionally, the reduced detection rates may reflect suboptimal model parameters or insufficient representative data for these diseases in the training set. These results could be improved by fine-tuning the model parameters or retraining the model using larger datasets specifically enriched with symptom profiles of HIV/AIDS and typhoid fever cases.

The interpretability provided by LIME proved essential in understanding how symptoms influenced each diagnosis,

thereby increasing model transparency. Identifying influential symptoms enabled insights into potential misclassifications, providing valuable context to healthcare workers. This interpretability supports clinical decision-making by allowing users to clearly understand the reasoning behind each diagnosis, thus enhancing system confidence.

The system significantly benefited from GPT-3.5's capability to generate natural language explanations. GPT-3.5 facilitated informed decision-making and improved user comprehension by translating complex model outputs into accessible explanations. Clear, comprehensible explanations could increase acceptance of AI-assisted diagnoses, especially in low-resource settings lacking specialized diagnostic tools and medical expertise. Integrating LIME and GPT-3.5, which effectively reduces the ML model's "black box" opacity, not only enhances disease diagnosis but also empowers healthcare workers by clarifying medical conditions and the rationale behind system-generated recommendations. To address the limitations associated with potential overgeneralizations or inaccuracies inherent in GPT-3.5 outputs, incorporating domain-specific fine-tuning and regular validation with expert feedback is recommended.

In conclusion, this study developed a mobile-based XAI system for diagnosing febrile diseases, integrating RF for disease prediction, LIME for interpretation, and GPT-3.5 for providing natural language explanations. This integration directly addressed the inherent "black box" nature of ML models, significantly improving transparency and trustwor-

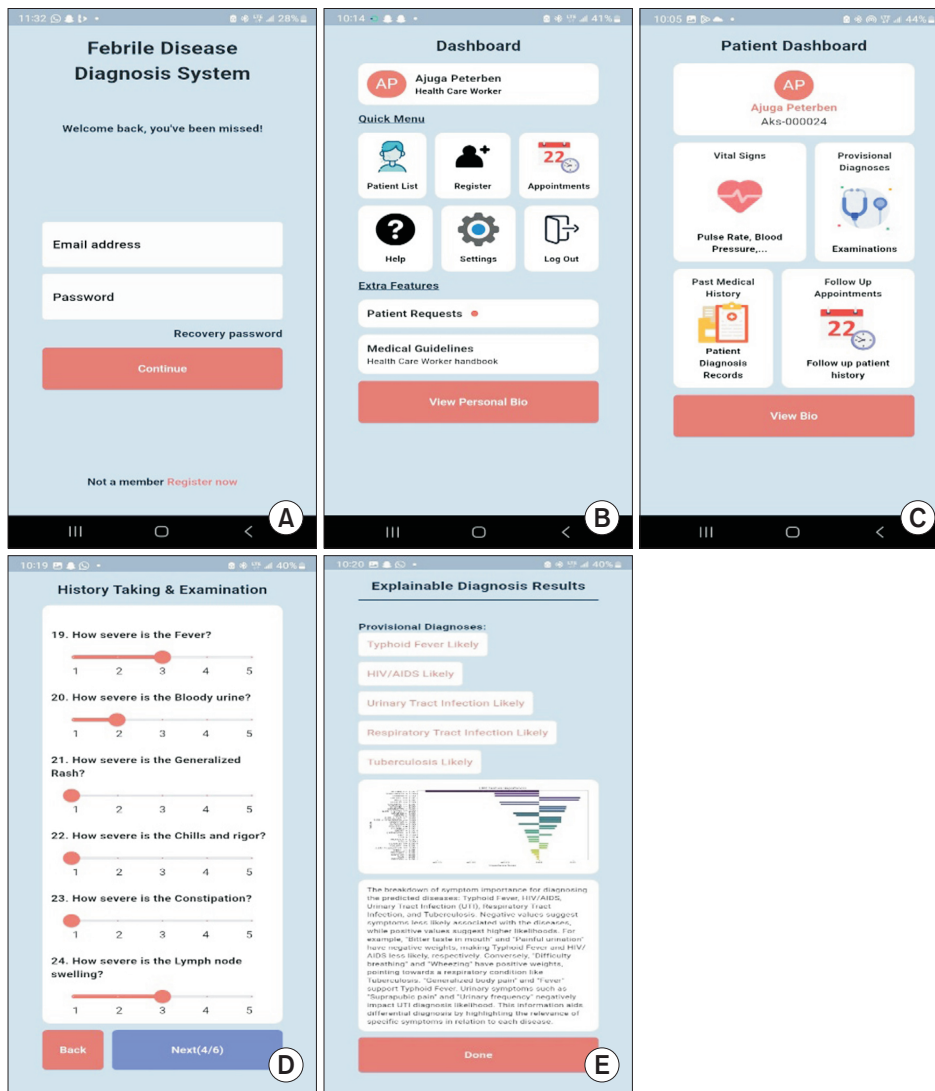


Figure 5. Mobile application for the febrile disease diagnostics system: (A) User Login, (B) Healthcare Worker, (c) Patient Account Dashboard, (D) History Taking and Examination, and (E) Explainable Diagnosis Results.

thiness in practical AI-driven healthcare diagnostics. The system achieved robust performance in diagnosing certain diseases, particularly malaria, with notably high precision, recall, and F1-scores. Additionally, the study demonstrated the potential of combining RF, LIME, and GPT-3.5 to create a user-friendly diagnostic tool accessible to both healthcare professionals and non-specialists in resource-constrained environments. Nevertheless, the study encountered some limitations. First, the system's diagnostic accuracy varied significantly across diseases, demonstrating particular challenges in detecting typhoid fever and HIV/AIDS due to lower precision and recall, thus necessitating further refinement. Second, the dataset utilized was limited to a specific population, potentially restricting the model's generalizability across diverse demographic and geographic contexts. Another notable limitation was the exclusion of pediatric patients (under 5 years of age), creating a gap in the system's

applicability for diagnosing young children.

Future research directions can address these limitations. Expanding the dataset to encompass more diverse populations and additional febrile diseases could increase the generalizability and robustness of the diagnostic model. Moreover, including data from pediatric patients and developing specialized diagnostic models tailored to this age group would significantly broaden the system's applicability. Real-time data updates and feedback mechanisms could further improve accuracy and adaptability. Additionally, future studies should incorporate structured human evaluations of the interpretability components, specifically LIME visualizations and GPT-generated explanations, utilizing Likert-scale assessments and obtaining expert feedback from medical professionals to validate their clinical relevance and usability.

Conflict of Interest

No potential conflict of interest relevant to this article was reported.

Acknowledgments

This study was supported by the New Frontier Research Fund (Grant No. NFRFE-201901365) awarded to Okure U. Obot, Christie Akwaowo, and Faith-Michael Uzoka.

ORCID

Kingsley F. Attai (<https://orcid.org/0000-0002-2199-5049>)
 Constance Amannah (<https://orcid.org/0000-0002-6395-4229>)
 Moses Ekpenyong (<https://orcid.org/0000-0001-6774-5259>)
 Daniel E. Asuquo (<https://orcid.org/0000-0002-0189-737X>)
 Oryina K. Akputu (<https://orcid.org/0000-0003-4520-0084>)
 Okure U. Obot (<https://orcid.org/0000-0001-7001-0114>)
 Peterben C. Ajuga (<https://orcid.org/0009-0009-9861-3940>)
 Jeremiah C. Obi (<https://orcid.org/0000-0001-9556-7314>)
 Omosivie Maduka (<https://orcid.org/0000-0002-8243-8131>)
 Christie Akwaowo (<https://orcid.org/0000-0001-5265-8267>)
 Faith-Michael Uzoka (<https://orcid.org/0000-0002-2414-5349>)

References

1. Attai K, Amannejad Y, Vahdat Pour M, Obot O, Uzoka FM. A systematic review of applications of machine learning and other soft computing techniques for the diagnosis of tropical diseases. *Trop Med Infect Dis* 2022; 7(12):398. <https://doi.org/10.3390/tropicalmed7120398>
2. Khan N, Okoli CN, Ekpin V, Attai K, Chukwudi N, Sabi H, et al. Adoption and utilization of medical decision support systems in the diagnosis of febrile diseases: a systematic literature review. *Expert Syst Appl* 2023;220:119638. <https://doi.org/10.1016/j.eswa.2023.119638>
3. Asuquo D, Attai K, Obot O, Ekpenyong M, Akwaowo C, Arnold K, et al. Febrile disease modeling and diagnosis system for optimizing medical decisions in resource-scarce settings. *Clinical eHealth* 2024;7:52-76. <https://doi.org/10.1016/j.cej.2024.05.001>
4. Kayode OA. Artificial intelligence (AI) and machine learning (ML) in health information management: its roles in clinical coding, data analytics, and workflow optimisation. *Afr J Educ Manag Teach Entrep Stud* 2024; 12(2):1-9.
5. Awotunde JB, Imoize AL, Salako DP, Farhaoui Y. An enhanced medical diagnosis system for malaria and typhoid fever using genetic neuro-fuzzy system. In: Farhaoui Y, Rocha A, Brahmia Z, Bhushab B, editors. *The international conference on artificial intelligence and smart environment*. Cham, Switzerland: Springer; 2023. p. 173-83. https://doi.org/10.1007/978-3-031-26254-8_25
6. Mariki M, Mkoba E, Mduma N. Combining clinical symptoms and patient features for malaria diagnosis: machine learning approach. *Appl Artif Intell* 2022;36(1):2031826. <https://doi.org/10.1080/08839514.2022.2031826>
7. Morales-Sanchez R, Montalvo S, Riano A, Martinez R, Velasco M. Early diagnosis of HIV cases by means of text mining and machine learning models on clinical notes. *Comput Biol Med* 2024;179:108830. <https://doi.org/10.1016/j.compbiomed.2024.108830>
8. Rajpurkar P, O'Connell C, Schechter A, Asnani N, Li J, Kiani A, et al. CheXaid: deep learning assistance for physician diagnosis of tuberculosis using chest x-rays in patients with HIV. *NPJ Digit Med* 2020;3:115. <https://doi.org/10.1038/s41746-020-00322-2>
9. Stokes K, Castaldo R, Franzese M, Salvatore M, Fico G, Pokvic LG, et al. A machine learning model for supporting symptom-based referral and diagnosis of bronchitis and pneumonia in limited resource settings. *Biocybern Biomed Eng* 2021;41(4):1288-302. <https://doi.org/10.1016/j.bbe.2021.09.002>
10. Su M, Guo J, Chen H, Huang J. Developing a machine learning prediction algorithm for early differentiation of urosepsis from urinary tract infection. *Clin Chem Lab Med* 2022;61(3):521-9. <https://doi.org/10.1515/cclm-2022-1006>
11. Attai K, Ekpenyong M, Amannah C, Asuquo D, Ajuga P, Obot O, et al. Enhancing the interpretability of malaria and typhoid diagnosis with explainable AI and large language models. *Trop Med Infect Dis* 2024;9(9):216. <https://doi.org/10.3390/tropicalmed9090216>
12. Tiwari R. Explainable AI (XAI) and its applications in building trust and understanding in AI decision making. *Int J Sci Res Eng Manag* 2023;7:1-13. <https://doi.org/10.55041/IJSREM17592>
13. Attai K, Akwaowo C, Asuquo D, Esubok NE, Nelson UA, Dan E, et al. Explainable AI modelling of comorbidity in pregnant women and children with tropical febrile conditions. *Proceedings of the International Conference on Artificial Intelligence and its Applications*; 2023 Nov 9-10; Mahebourg, Mauritius. p. 152-9. <https://doi.org/10.4258/hir.2025.31.2.125>

- doi.org/10.59200/ICARTI.2023.022
14. Onyekwe H, Ekerete I, Garcia-Constantino M, Nugent C. Modelling activities of daily living using local interpretable model-agnostic explanation algorithm. Proceedings of 2024 35th Irish Signals and Systems Conference (ISSC); 2024 Jun 13-14; Belfast, UK. p. 1-6. <https://doi.org/10.1109/ISSC61953.2024.10602854>
15. Yenduri G, Ramalingam M, Selvi GC, Supriya Y, Srivastava G, Maddikunta PK, et al. GPT (generative pre-trained transformer): a comprehensive review on enabling technologies, potential applications, emerging challenges, and future directions. IEEE Access 2024;12:54608-49. <https://doi.org/10.1109/ACCESS.2024.3389497>
16. University of Uyo Teaching Hospital; Mount Royal University. NFRF project patient dataset with febrile diseases (dataset) [Internet]. Geneva, Switzerland: Zenodo; 2024 [cited at 2025 Apr 27]. Available from: <https://doi.org/10.5281/zenodo.13756418>.
17. Ghosh D, Cabrera J. Enriched Random Forest for High Dimensional Genomic Data. IEEE/ACM Trans Comput Biol Bioinform 2022;19(5):2817-28. <https://doi.org/10.1109/TCBB.2021.3089417>
18. Anbarasi MS, Janani V. Ensemble classifier with random forest algorithm to deal with imbalanced healthcare data. Proceedings of 2017 International Conference on Information Communication and Embedded Systems (ICICES); 2017 Feb 23-24; Chennai, India. p. 1-7. <https://doi.org/10.1109/ICICES.2017.8070752>